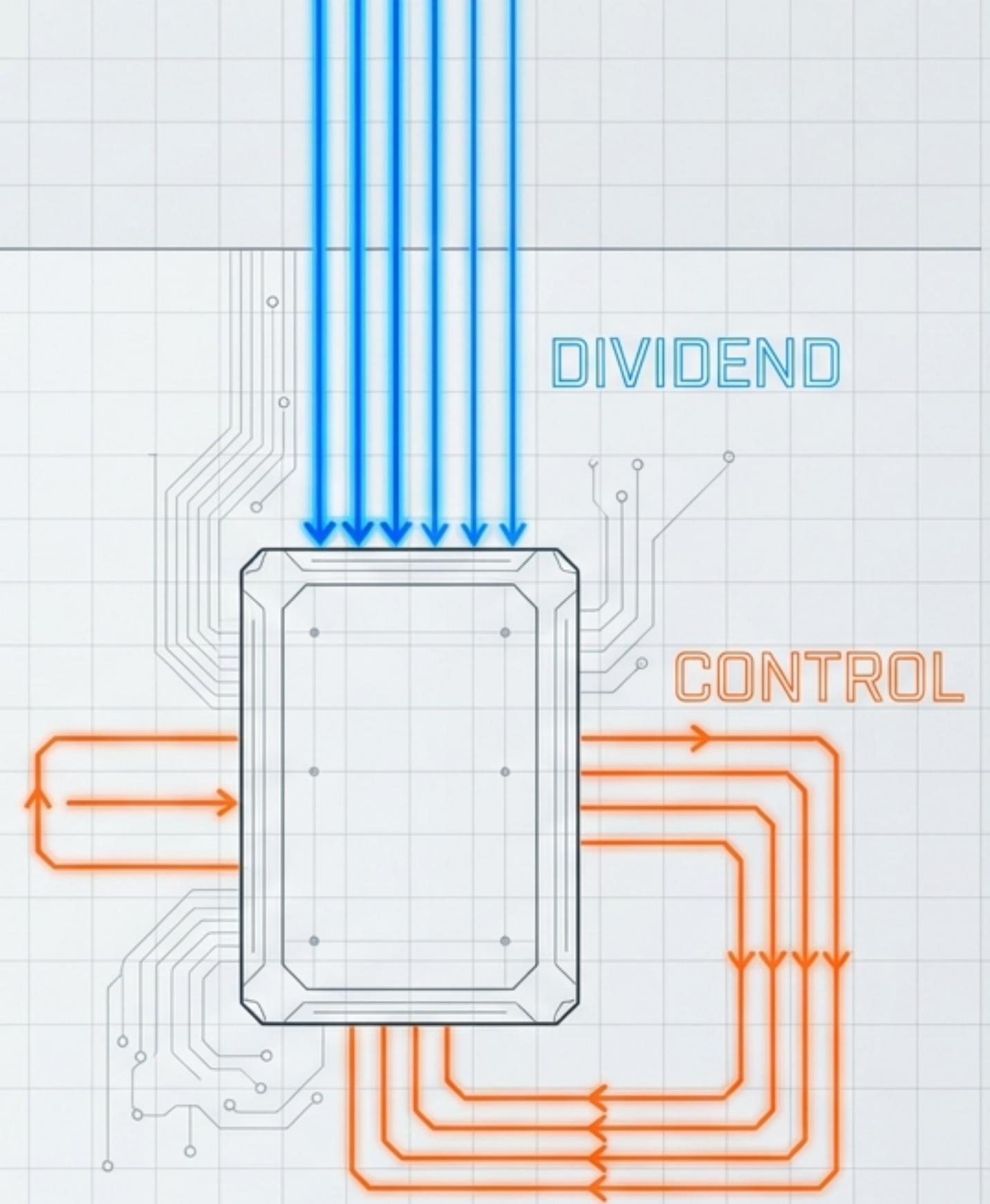


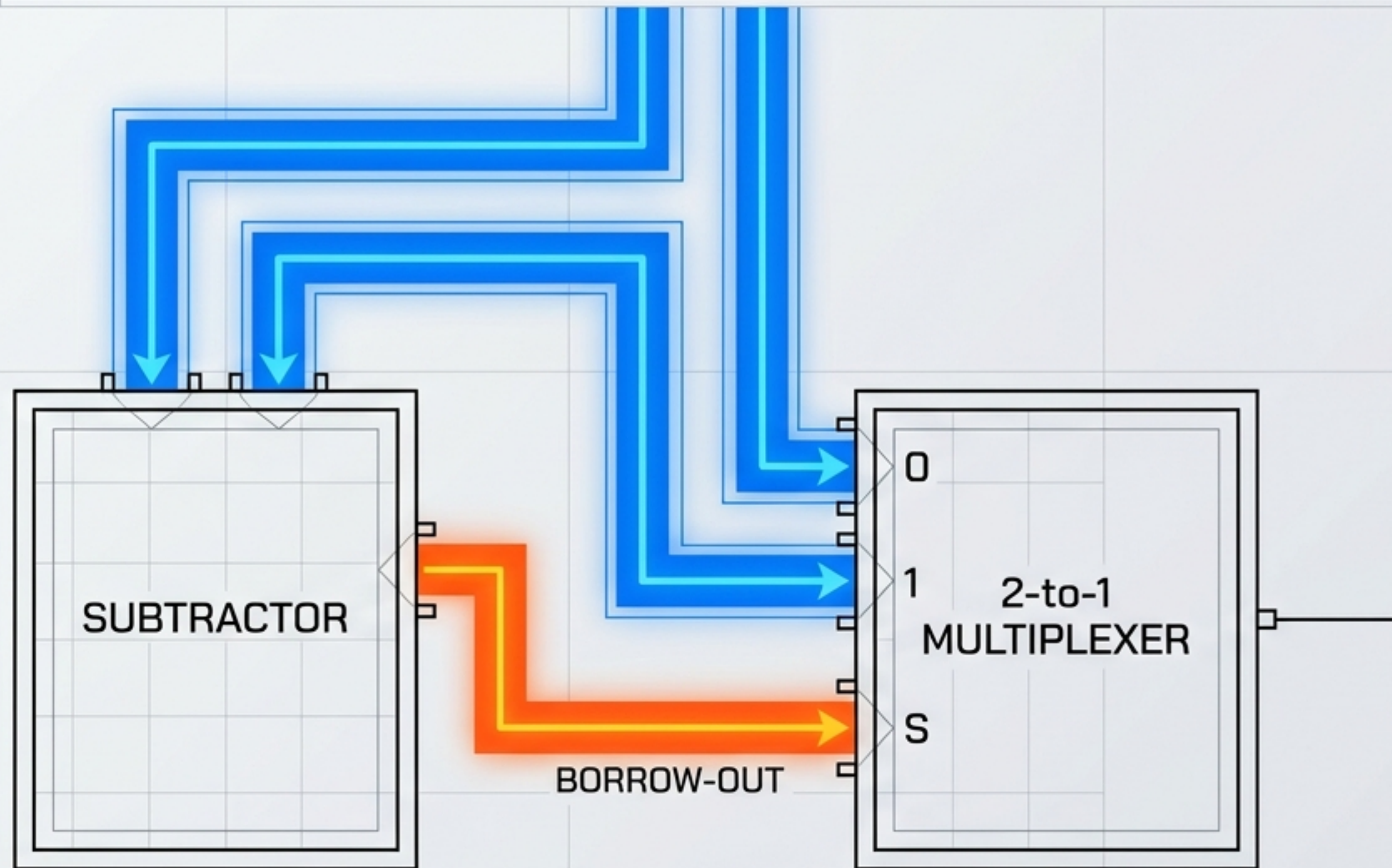
COMBINATIONAL DIVIDERS

HARDWARE ARRAY FOR BINARY DIVISION

$$\text{DIVIDEND} = \text{DIVISOR} * \text{QUOTIENT} + \text{REMAINDER}$$



UNIT CELL ARCHITECTURE



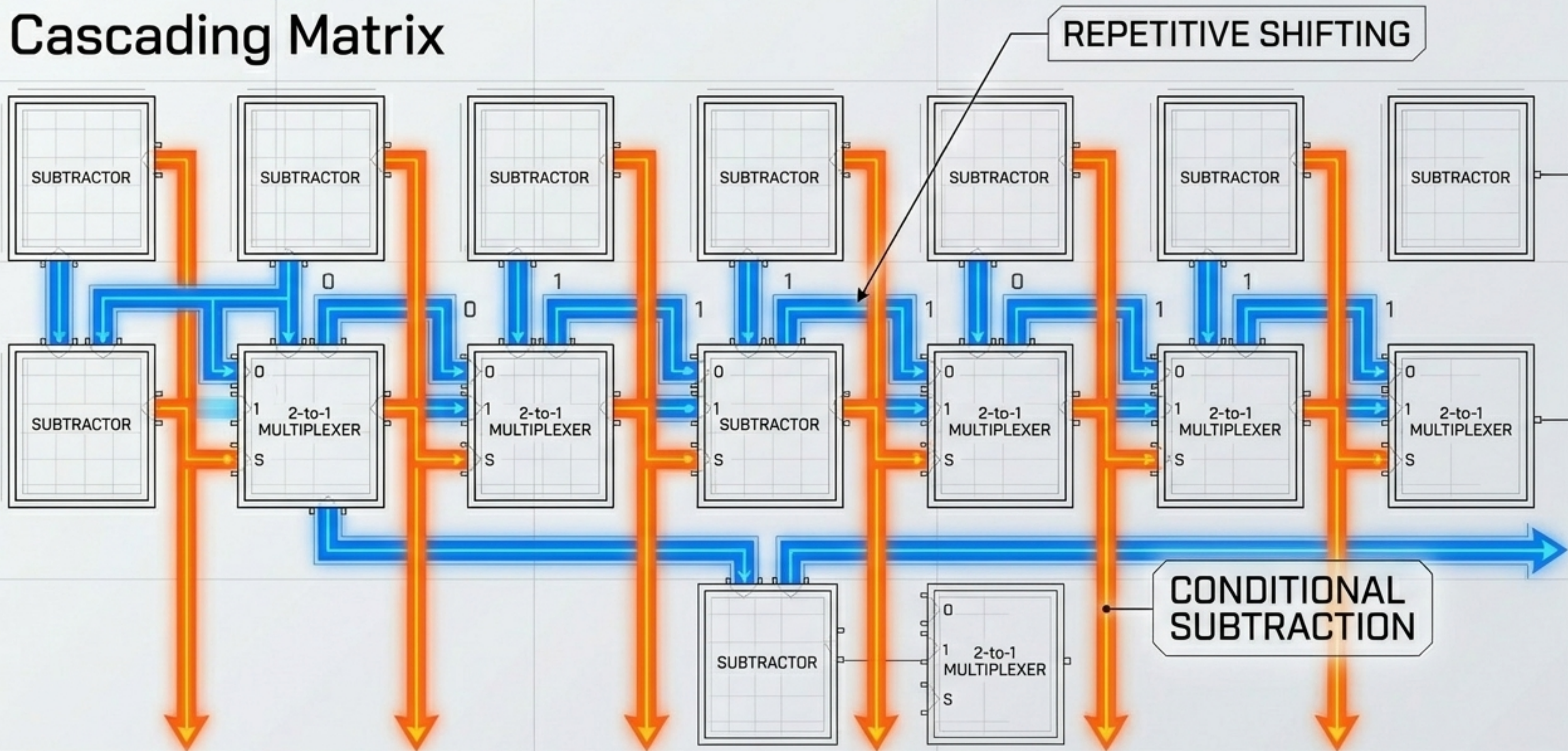
RULE:
BORROW-OUT
CONTROLS THE
MUX

PATH 0:
PASS SUBTRAHEND
(IF BORROW = 0)

PATH 1:
PASS DIFFERENCE
(IF BORROW = 1)

THE ARRAY STRUCTURE

Cascading Matrix



STEP-BY-STEP ANALOGY

CONCEPT MAPPING TABLE		
DIVIDEND	=	TOTAL CHOCOLATES
DIVISOR	=	NUMBER OF CHILDREN
QUOTIENT	=	NUMBER OF SUCCESSFUL GROUPS
REMAINDER	=	LEFTOVERS

20 CHOCOLATES DIVIDED BY 4 CHILDREN

- 1
- STEP 1: CHECK IF 4 CHOCOLATES CAN BE GIVEN.
- 2
- STEP 2: SUBTRACT 4.
- 3
- STEP 3: NUMBER OF SUCCESSFUL GROUPS = QUOTIENT.
- 4
- STEP 4: LEFTOVERS = REMAINDER.